

Digital Faraday: Sovereign Comms Blueprint

A technical guide to mesh networking and local data sovereignty.

1. Meshtastic Performance Benchmarks

Environment	Effective Range
Urban / High Interference	0.5 - 2 miles
Rural / Line of Sight	10 - 20 miles
Mountain Top / High Gain	50+ miles (Projected)

2. Self-Hosted NAS Sizing Formula

*Required Capacity = ((Users * 50GB Photos) + (Users * 200GB Video)) * 1.5 Redundancy Multiplier*

3. Hardware Recommendations

- Base Station: Raspberry Pi 5 (8GB)
- Storage: 8TB Enterprise HDD (Exos/IronWolf)
- RAID: RAID 1 (Mirroring) for data integrity